

Chess Rating Estimation — Progress Notes

October 7, 2026

1 Setting

Classify the rating bin of a game from the game itself (moves, clocks, opening, time control). One row per game, label `elo_bin` from the average Elo of both players. Data: Lichess standard rated rapid games, April–June 2017. Model: XGBoost. Headline metric: accuracy; per-bin accuracy and the confusion matrix as diagnostics.

2 Sampling

34.6M games read, 7.64M rapid, 4.53M eligible. Filters: rated, rapid ($480 \leq \text{base} + 40 \cdot \text{inc} \leq 1499$), no bots, no abandoned or rules-infraction games, $|\Delta\text{Elo}| \leq 100$, at least 20 half-moves. Stratified reservoir sampling over 9 bins of 200 Elo, cap 50 000 per bin (333 478 games; 13 h single-threaded).

Bin	Kept	Eligible	Retention
< 800	47	47	100 %
[800, 1000)	22 340	22 340	100 %
[1000, 1200)	50 000	232 583	21.5 %
[1200, 1400)	50 000	762 977	6.5 %
[1400, 1600)	50 000	1 383 785	3.6 %
[1600, 1800)	50 000	1 329 333	3.8 %
[1800, 2000)	50 000	662 901	7.5 %
[2000, 2200)	50 000	121 233	41.2 %
≥ 2200	11 091	11 091	100 %

The tails are exhausted and stay below the target. For modelling the bins are merged to 7 classes: < 1000 (from < 800 and $[800, 1000)$) and ≥ 2000 (from $[2000, 2200)$ and ≥ 2200).

3 Train/test split

The label is mostly a property of the player, so train and test must be player-disjoint; otherwise the model can recognise players by their play. A game whose players land on different sides is dropped.

- 91 620 players; 31% have one game, but players with ≥ 10 games hold 68% of all player slots. 39% of players span more than one bin.
- Player graph (edge = played each other): 97.3% of players and 99.5% of games are in one connected component, so a split cannot follow components (Fig. 1).

- With independent assignment and fractions p_i , a game survives with probability $\sum_i p_i^2$ and lands in split i with probability p_i^2 . An 80/10/10 player split gives a 97/1.5/1.5 game split and loses 34%. Graph partitioning could lose less; not measured.

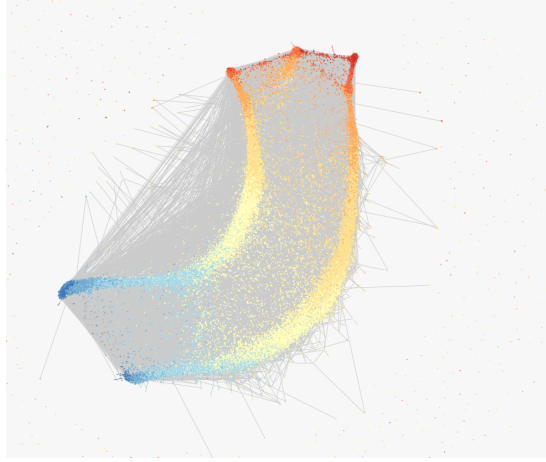


Figure 1: Player graph, nodes coloured by rating bin.

Test size. The accuracy measured on the test set differs from the true accuracy by chance, because the test games are a random selection. For n test games and accuracy p , the standard error is

$$SE = \sqrt{\frac{p(1-p)}{n}}.$$

Since $p(1-p) \leq 0.25$ for any p , the worst case is

$$SE \leq \frac{0.5}{\sqrt{n}},$$

which needs no assumption about the model. The 95% interval is $\pm 1.96 \cdot SE$.

With independent assignment, test games = $p_{\text{test}}^2 \cdot 333\,478$ and train games = $(1 - p_{\text{test}})^2 \cdot 333\,478$, so a larger test set costs training data:

Test games	p_{test}	Train games	Dropped	95% interval
5 000	0.122	$\approx 257\,000$	21%	± 1.4 pp
10 000	0.173	$\approx 228\,000$	29%	± 1.0 pp
20 000	0.245	$\approx 190\,000$	37%	± 0.7 pp
50 000	0.387	$\approx 125\,000$	47%	± 0.4 pp

At 10 000 test games the interval is about ± 1 pp while the training set stays large; doubling the test set would narrow it by 0.3 pp at the cost of about 38 000 training games. Hence $p_{\text{test}} \approx 0.173$.

The formula treats test games as independent. Games by the same player are correlated, so the true interval is wider by an amount not yet known. It can be measured once predictions exist, by resampling test players rather than games. Until then, ± 1.0 pp is a lower bound on the uncertainty.

Result. Independent assignment within 9 strata (quantiles of player mean Elo), seed 42, train/test = 0.827/0.173.

	Games	Players	Share of kept
Train	229 402	71 198	96.0%
Test	9 666	7 688	4.0%
Dropped	94 410 (28.3%)		

Retention 71.7% (predicted 71.4%). Disjointness check passes. Test ratings are 7–35 Elo lower than train across the deciles (mean 1563 vs. 1578; KS 0.023).

Games per class after merging:

	<1000	1000–1200	1200–1400	1400–1600	1600–1800	1800–2000	≥ 2000
Train	14 851	34 545	34 514	34 430	34 389	33 862	42 811
Test	778	1 449	1 421	1 417	1 416	1 576	1 609

Baselines (test). Majority class (≥ 2000): 16.6%. Uniform: 14.3%. Every model accuracy is reported next to these.

Validation. No fixed validation set (it would pay the p_i^2 cost again). Hyperparameters use 5-fold CV inside train with folds assigned by player. A game is a validation row of fold k if both players are in k , a fit row if neither is, and is used for fitting in the other folds if its players are in different folds. Each validation fold then holds only about 1/25 of the games ($\approx 9\,000$).

Rejected.

- Temporal split: same players on both sides, leaks the label.
- Random 80/10/10 player split: see above.
- Split the eligible pool first, then sample: protects the tails, but means redoing the 13h sampling run.

4 Features

Excluded (label or identity): `white_elo`, `black_elo`, `avg_elo`, `rating_diff`, `white`, `black`, `game_id`, `utc_date`, `utc_time`, `split`. Current inputs: `result`, `termination`, `half-moves`, `moves`, `clocks`, `opening_family`, `tc_base`, `tc_increment`. Anything fitted on data (e.g. opening target encoding) is fitted on training rows only, inside each CV fold.

Stockfish (in progress). Engine features (centipawn loss, blunders, best-move agreement) for all train/test games, run on a rented machine with a hard cap of \$20. Calibration on 350 games (50 per bin), depths 15/18/20 and MultiPV 1–3: per-game ACPL vs. Elo has Spearman $\rho \approx -0.52$ for every configuration, with overlapping 95% CIs, so no measurable gain from more depth or MultiPV. Depth 15, MultiPV 1 is the leading candidate (est. \$5, unverified cloud price); final choice open.

5 Open

- Engine depth/MultiPV decision and the full remote run.
- Remaining PGN features (clock usage, development, castling).
- Class weights for the sparse < 1000 class.
- CV fold construction.
- Baseline XGBoost, then lift from engine features.